

CS5494



香港城市大學
City University of Hong Kong
Innovating into the Future

Welcome to Topics in Generative AI

Dr. Jianyuan Guo
Fall 2026, Dept. CS, CityU

Logistics



Instructor

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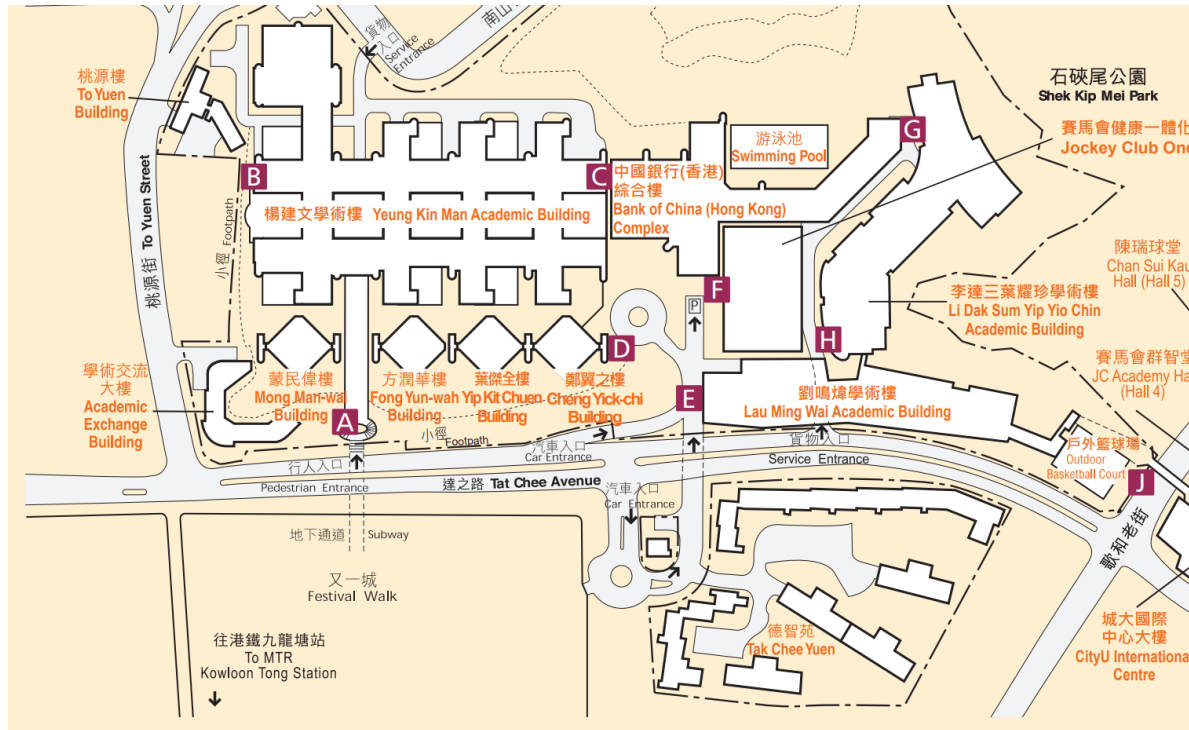
Reference book

- David Foster, “Generative Deep Learning: Teaching Machines To Paint, Write, Compose, and Play (2nd Ed.)”, O'Reilly, 2023.
 - https://github.com/davidADSP/Generative_Deep_Learning_2nd_Edition
- Omar Sanseviero, Pedro Cuenca, Apolinário Passos, Jonathan Whitaker, “Hands-On Generative AI with Transformers and Diffusion Models”, O'Reilly, 2024. 3
- “Basic Generative AI: Beginner's Guide to Artificial Intelligence, ChatGPT and Machine Learning, Practical AI Applications”, Freeman Publishing, 2024.

We will use PyTorch in labs. PyTorch documentation can be found at <https://pytorch.org/docs/stable/index.html>

Lectures and Tutorials

- Tutorial:
 - Monday, 18:00 – 18:50, Yeung Kin Man Acad Building B7701
 - Monday, 19:00 – 19:50, Yeung Kin Man Acad Building B7520
- Lecture:
 - Monday, 20:00 – 21:50, Yeung Kin Man Acad Building LT-17



What to Learn?

- Week 1: Introduction Perception vs. Generation
- Week 2: Basic of probability distributions
- Week 3: AR model
- Week 4: LLM
- Week 5: GAN model
- Week 6: VAE model
- Week 7: Diffusion model
- Week 8: Agent
- Week 9: 3D generation
- Week 10: Advanced Topics
- Week 11-13: Course Presentations

Course Overview



Style

- Lecture + Tutorial (coding on Google colab) + assignment + project (including week11-13 presentations) + final exam.

Grading

- Final Examination: 40%
- Assignment: 20% (published at week 3, due in week 7)
 - Groups of 1-3 students
- Project: 40% (published at week 6, due in week 13)
 - Groups of 1-3 students
 - Report: 30%
 - Presentation: 10%

Communication and Policies



Communication

- Canvas Discussions
- Email
- It is your own responsibility to check Canvas and University e-mail account regularly for announcements and updates.

If your performance on assessments is affected by illness or misadventure

- Follow proper bureaucratic procedures, e.g., have professional practitioner signed doc
- notify coordinator by email as soon as anything begins to go wrong
- there is a similar process if you need special arrangements, e.g., for religious observance, military service, representative sports

Work division

- Each team member should do similar amount of work
- More contribution, better grade

Late Submission in CS5494



Suppose you hand in work after the deadline:

- If you have not been granted special consideration or arrangements
 - A penalty of 5% of the maximum marks will be taken per day (or part) late. After ten days, you will be awarded a mark of zero.
- e.g. If an assignment is worth 40% of the final mark and you are one hour late submitting, then the maximum marks possible would be 38%.
- e.g. If an assignment is worth 40% of the final mark and you are 28 hours late submitting, then the maximum marks possible marks would be 36%.
- Warning: Google cloud GPUs (quotas) may be **not enough** near deadlines
- Submit early; you can resubmit if there is time before the deadline



Thank You!